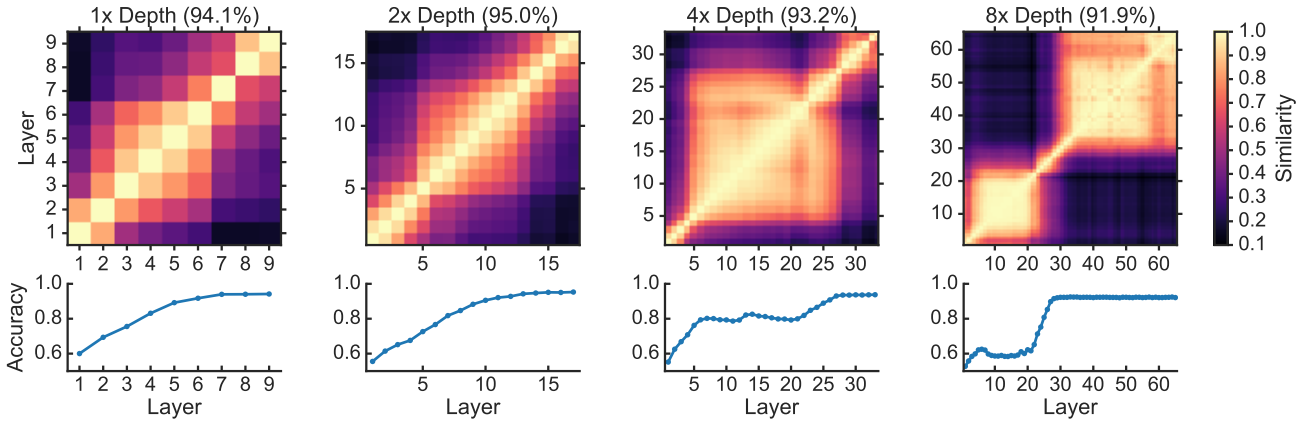

Figure 3. CKA reveals when depth becomes pathological. **Top**: Linear CKA between layers of individual networks of different depths on the CIFAR-10 test set. Titles show accuracy of each network. Later layers of the 8x depth network are similar to the last layer. **Bottom**: Accuracy of a logistic regression classifier trained on layers of the same networks is consistent with CKA.

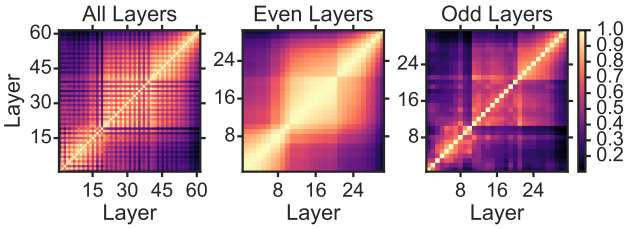

Figure 4. Linear CKA between layers of a ResNet-62 model on the CIFAR-10 test set. The grid pattern in the left panel arises from the architecture. Right panels show similarity separately for even layer (post-residual) and odd layer (block interior) activations. Layers in the same block group (*i.e.* at the same feature map scale) are more similar than layers in different block groups.

network, accuracy plateaus less than halfway through the network. When applied to ResNets (He et al., 2016), CKA reveals no pathology (Figure 4). We instead observe a grid pattern that originates from the architecture: Post-residual activations are similar to other post-residual activations, but activations within blocks are not.

CKA is equally effective at revealing relationships between layers of different architectures. Figure 5 shows the relationship between different layers of networks with and without residual connections. CKA indicates that, as networks are made deeper, the new layers are effectively inserted in between the old layers. Other similarity indexes fail to reveal meaningful relationships between different architectures, as we show in Appendix F.5.

In Figure 6, we show CKA between networks with different layer widths. Like Morcos et al. (2018), we find that increasing layer width leads to more similar representations between networks. As width increases, CKA approaches 1; CKA of earlier layers saturates faster than later layers. Networks are generally more similar to other networks of the same width than they are to the widest network we trained.

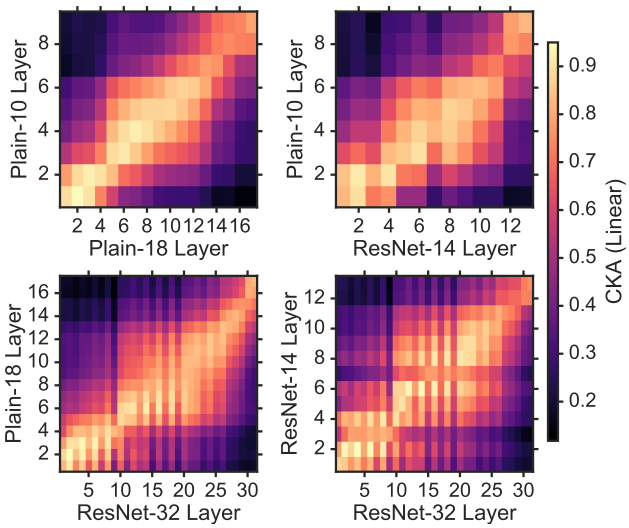

Figure 5. Linear CKA between layers of networks with different architectures on the CIFAR-10 test set.

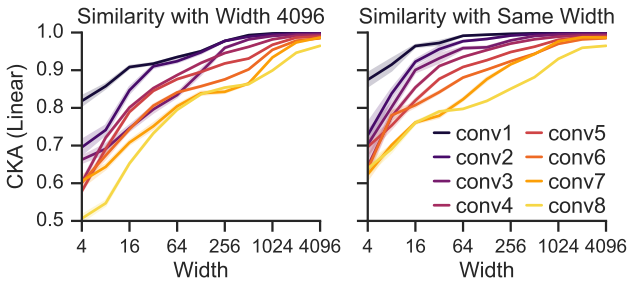

Figure 6. Layers become more similar to each other and to wide networks as width increases, but similarity of earlier layers saturates first. **Left**: Similarity of networks with the widest network we trained. **Middle**: Similarity of networks with other networks of the same width trained from random initialization. All CKA values are computed between 10 networks on the CIFAR-10 test set; shaded regions reflect jackknife standard error.

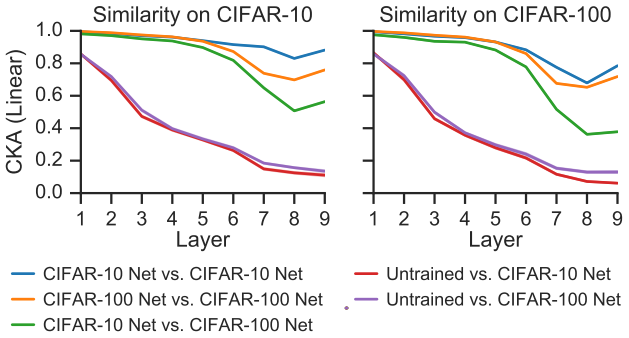

Figure 7. CKA shows that models trained on different datasets (CIFAR-10 and CIFAR-100) develop similar representations, and these representations differ from untrained models. The left panel shows similarity between the same layer of different models on the CIFAR-10 test set, while the right panel shows similarity computed on CIFAR-100 test set. CKA is averaged over 10 models of each type (45 pairs).

6.3. Similar Representations Across Datasets

CKA can also be used to compare networks trained on different datasets. In Figure 7, we show that models trained on CIFAR-10 and CIFAR-100 develop similar representations in their early layers. These representations require training; similarity with untrained networks is much lower. We further explore similarity between layers of untrained networks in Appendix F.3.

6.4. Analysis of the Shared Subspace

Equation 14 suggests a way to further elucidating what CKA is measuring, based on the action of one representational similarity matrix (RSM) YY^{T} applied to the eigenvectors $\mathbf{u}_X^i$ of the other RSM XX^{T}. By definition, $XX^{\mathrm{T}}\mathbf{u}_X^i$ points in the same direction as $\mathbf{u}_X^i$, and its norm $||XX^{\mathrm{T}}\mathbf{u}_X^i||_2$ is the corresponding eigenvalue. The degree of scaling and rotation by YY^{T} thus indicates how similar the action of YY^{T} is to XX^{T}, for each eigenvector of XX^{T}. For visualization purposes, this approach is somewhat less useful than the CKA summary statistic, since it does not collapse the similarity to a single number, but it provides a more complete picture of what CKA measures. Figure 8 shows that, for large eigenvectors, XX^{T} and YY^{T} have similar actions, but the rank of the subspace where this holds is substantially lower than the dimensionality of the activations. In the penultimate (global average pooling) layer, the dimensionality of the shared subspace is approximately 10, which is the number of classes in the CIFAR-10 dataset.

7. Conclusion and Future Work

Measuring similarity between the representations learned by neural networks is an ill-defined problem, since it is not entirely clear what aspects of the representation a similarity index should focus on. Previous work has suggested that there is little similarity between intermediate layers of neural networks trained from different random initializations (Raghu et al., 2017; Wang et al., 2018). We propose CKA as a method for comparing representations of neural networks, and show that it consistently identifies correspondences between layers, not only in the same network trained from different initializations, but across entirely different architectures, whereas other methods do not. We also provide a unified framework for understanding the space of similarity indexes, as well as an empirical framework for evaluation.

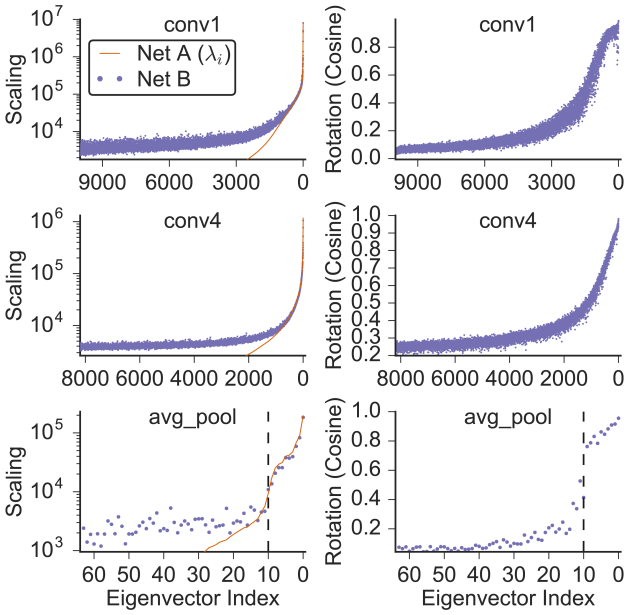

Figure 8. The shared subspace of two networks trained on CIFAR-10 from different random initializations is spanned primarily by the eigenvectors corresponding to the largest eigenvalues. Each row represents a different network layer. Note that the average pooling layer has only 64 units. **Left**: Scaling of the eigenvectors $\mathbf{u}_X^i$ of the RSM XX^{T} from network A by RSMs of networks A and B. Orange lines show $||XX^{\mathrm{T}}\mathbf{u}_X^i||_2$, *i.e.* the eigenvalues. Purple dots show $||YY^{\mathrm{T}}\mathbf{u}_X^i||_2$, the scaling of the eigenvectors of the RSM of network A by the RSM of network B. **Right**: Cosine of the rotation by the RSM of network B, $(\mathbf{u}_X^i)^{\mathrm{T}}YY^{\mathrm{T}}\mathbf{u}_X^i/||YY^{\mathrm{T}}\mathbf{u}_X^i||_2$.

We show that CKA captures intuitive notions of similarity, *i.e.* that neural networks trained from different initializations should be similar to each other. However, it remains an open question whether there exist kernels beyond the linear and RBF kernels that would be better for analyzing neural network representations. Moreover, there are other potential choices of weighting in Equation 14 that may be more appropriate in certain settings. We leave these questions as future work. Nevertheless, CKA seems to be much better than previous methods at finding correspondences between the learned representations in hidden layers of neural networks.

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